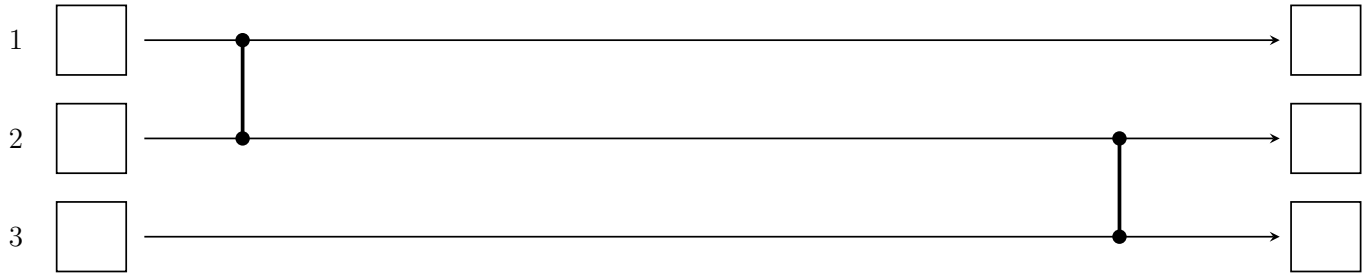
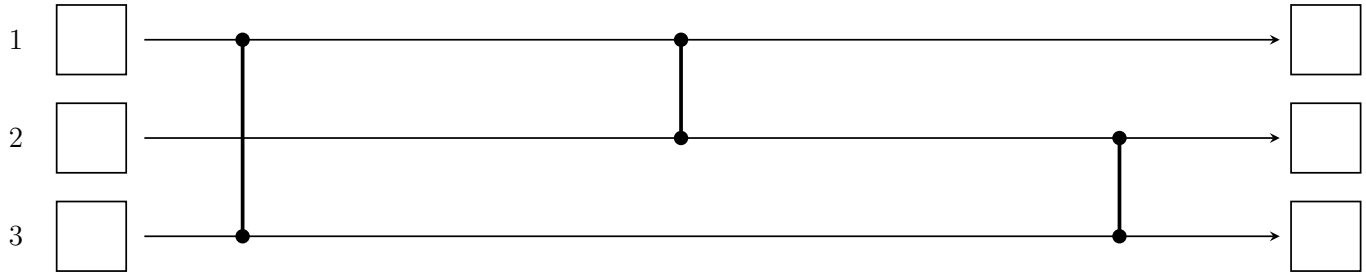
Move from left to right. At each bar, compare only the two lanes with black dots: the smaller value goes to the upper lane and the larger to the lower lane. Ties may stay put. Keep every bar fixed for every start; move cards only at bars. A machine sorts when the final values never get smaller from lane 1 down.

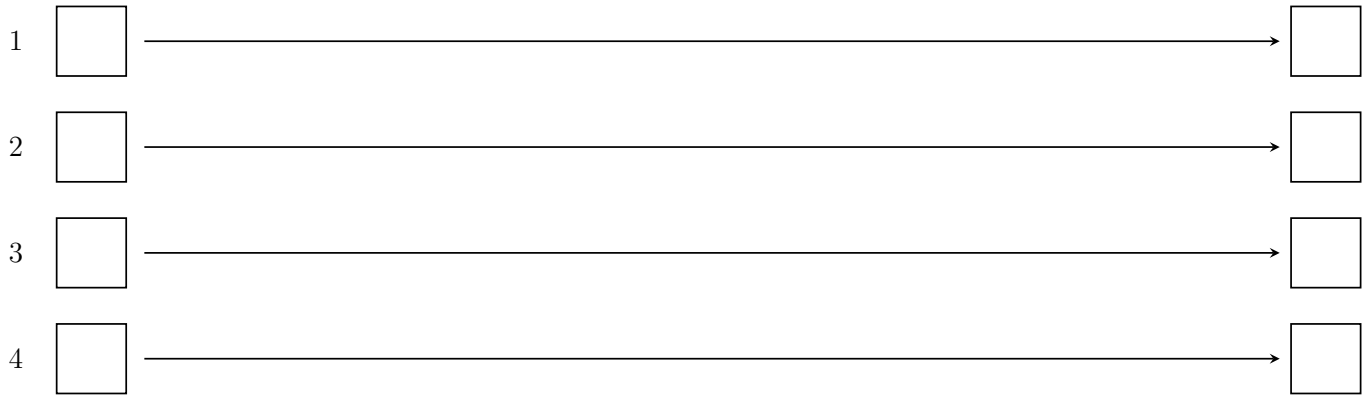
Problem 1: For each machine, decide whether it sorts every order of 1, 2, 3. Give a failing start or a complete test record. Decide whether either machine can fail with 1, 1, 3 or 1, 3, 3.



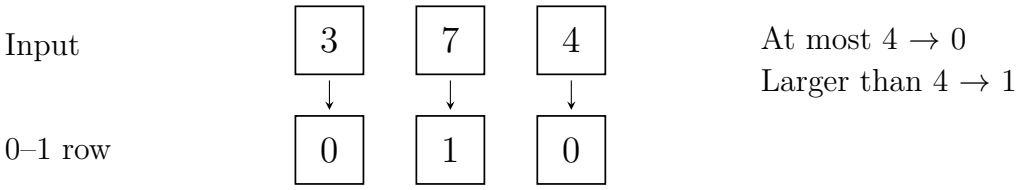


Problem 2: Build a four-lane machine that sorts every order of 1, 2, 3, 4. Use as few bars as you can.





Example: choose $t = 4$. Keep each card in its position.



Problem 3: Choose a number t. Replace a value by 0 when it is at most t, and by 1 when it is larger than t. For each start, find every different row of 0s and 1s that choosing t can produce.

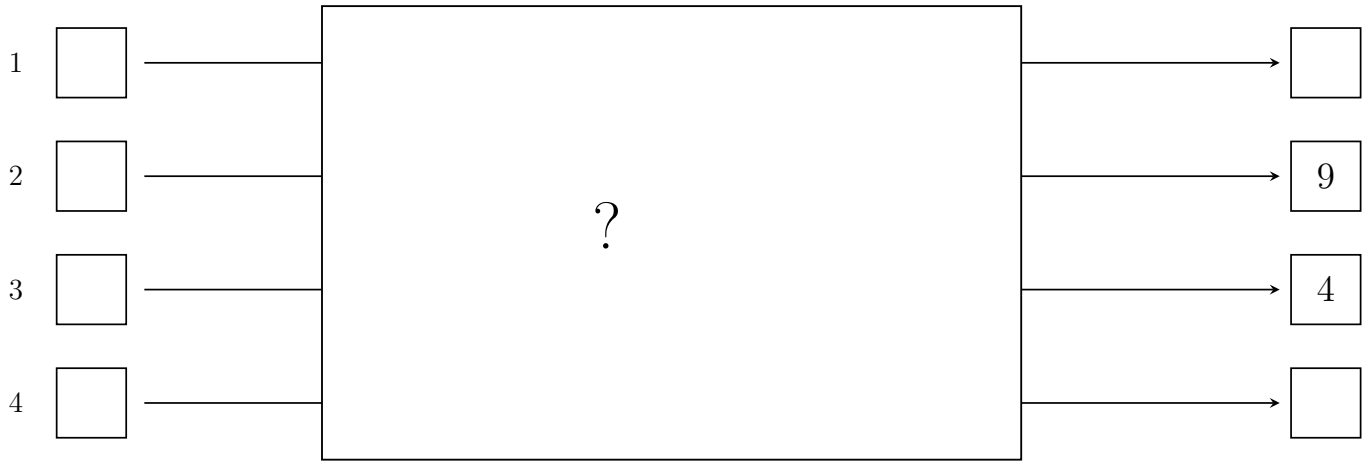
-2	8	5	8

6	2	9	4

4	4	4	4

Can replacing values before a bar ever give different final 0s and 1s from replacing them after the bar? Explain for any two values and any t.

Problem 4: A fixed four-lane machine sorts every start made of 0s and 1s. Could some other start finish with 9 in lane 2 and 4 in lane 3? Explain. Decide whether sorting every 0–1 start guarantees sorting every start made of numbers, including repeats.



Problem 5: Decide which machine sorts every start made of numbers. Give an explanation covering all inputs, including repeated values.

